

Longest Increasing Subsequence
Dynamic Programming II
(ADA-2023)

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- **Input:** A sequence of n integers $X = \langle x_1, \dots, x_n \rangle$.
- **Goal:** Find a subsequence Y of X such that Y is increasing and Y has maximum number of elements from X .
- **Example:**

x_1	x_2	x_3	x_4	x_5	x_6
11	14	13	7	8	15

$x_2 \quad x_5 \quad x_6$

14 8 15

$y_1 \quad y_2 \quad y_3$

11 14 15

An increasing subsequence x_1, x_2, x_6

11 13 15

$x_1 \quad x_3 \quad x_6$

- **Input:** A sequence of n integers $X = \langle x_1, \dots, x_n \rangle$.
- **Goal:** Find a subsequence Y of X such that Y is increasing and Y has maximum number of elements from X .

11 14 13 7 8 15

14 8 15

- **Input:** A sequence of n integers $X = \langle x_1, \dots, x_n \rangle$.
- **Goal:** Find a subsequence Y of X such that Y is increasing and Y has maximum number of elements from X .
- It must be that $Y = \langle \underline{x_{\varphi(1)}}, \underline{x_{\varphi(2)}}, \dots, \underline{x_{\varphi(k)}} \rangle$ such that $x_{\varphi(i)} < x_{\varphi(i+1)}$.
- There are two cases.
- **Case 1:** x_n is not in an optimal subsequence.

Recursively solve for $\{x_1, x_2, \dots, x_{n-1}\}$.

- **Case 2:** x_n is in an optimal subsequence.

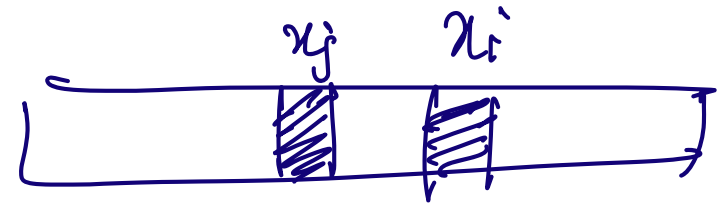
Longest increasing subsequence must contain x_n .

What are the subproblems?

$$\langle x_1, x_2, \dots, x_i \rangle$$

- Define subproblems over $X_i = \langle \underline{x_1, \dots, x_i} \rangle$ and $X = X_n$.
- $LS(x_i)$ is the length of a longest subsequence of X_i that contains x_i .
- $LIS(x_i)$ is the length of a longest subsequence of X_i .
- **Case 2:** x_i is in an optimal subsequence of X_i .

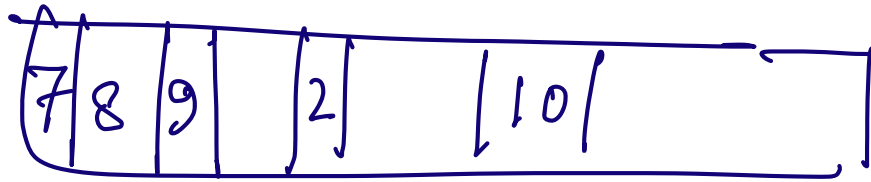
$$LS(x_i) = 1 + \max_{j < i: x_j < x_i} LS(x_j).$$



- Finally $LIS(x_n) = \max_{1 \leq k \leq n} LS(x_k)$.

$$LIS(x_i) = \max_{1 \leq j \leq i} LS(x_j)$$

Algorithm outline



- For every $1 \leq i \leq n$, initialize $\text{Length}(i) = 1$.
- Then, for every $i = 1, \dots, n$ and for $j = i + 1, \dots, n$.
 - If $x_j > x_i$, then
 $\text{Length}(j) = \max\{\text{Length}(j), \text{Length}(i) + 1\}$.
- Output $\max_{1 \leq i \leq n} \text{Length}(i)$.

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for ( $i = 1, \dots, n$ ) {  
    for ( $j = i + 1, i + 2, \dots, n$ )  
         $\text{Length}(j) = \max \begin{bmatrix} \text{Length}(j) \\ \text{Length}(i) + 1 \end{bmatrix}$   
    }  
}

# References

- Jeff Erickson Lecture Notes Section 3.6.
- Another lecture note is also attached.

